

Houssam Zenati

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RESEARCH INTERESTS	Statistical machine learning for systems whose decisions shape the data they subsequently learn from, especially when valid estimation requires nuisance functions, latent variables, or learned representations. Interests include nuisance-robust inference, adaptive experiments, distributional causal inference for structured outcomes, functionals of solutions to inverse and nested problems, policy learning, and applications to biology and biomedicine.
PROFESSIONAL EXPERIENCE	Gatsby Computational Neuroscience Unit, University College London, United Kingdom <i>Research Fellow</i> 2025–present Developing methods for semiparametric efficient distributional causal inference, adaptive data collection, and sequential decision-making. INRIA MIND / PreMeDical, France <i>Postdoctoral Researcher</i> 2023–2024 Research on causal inference and mediation analysis, including double machine learning estimators and open-source software development. Criteo AI Lab, France <i>PhD Researcher</i> 2020–2023 <i>Research Engineer</i> 2019–2020 Worked on counterfactual policy learning and online/bandit problems, including optimization methods, contextual bandits, and kernel methods. Institute for Infocomm Research (ASTAR), Singapore <i>Research Intern / Research Experience</i> 2017–2018 Worked on deep representation learning, generative adversarial networks, anomaly detection, and semi-supervised learning, with applications to medical imaging.
EDUCATION	Université Grenoble Alpes / INRIA Thoth, France 2023 Ph.D. in applied mathematics and machine learning. Thesis: <i>Efficient Methods in Counterfactual Policy Learning and Sequential Decision Making</i>. ENS Paris-Saclay / MVA, France 2019 M.Sc. in mathematics, vision, and learning. CentraleSupélec, France 2015–2019 M.Eng. in applied mathematics and computer science.
SELECTED PUBLICATIONS	Z. Shen*, H. Zenati*, N. Kallus, A. Gretton, K. Khamaru, A. Bibaut (2026). <i>Efficient Inference after Directionally Stable Adaptive Experiments</i>. Preprint. arXiv:2602.21478 H. Zenati, B. Bozkurt, A. Gretton (2026). <i>Kernel Treatment Effects with Adaptively Collected Data</i>. AISTATS. arXiv:2510.10245 H. Zenati, A. Gretton (2026). <i>Semiparametric Efficient Test for Interpretable Distributional Treatment Effects</i>. Preprint. arXiv:2605.08034 Girard, Samuel, Nathan Kallus, Jill-Jënn Vie, Arthur Gretton, Aurélien Bibaut, and Houssam Zenati (2026). <i>Fast Best-in-Class Regret for Contextual Bandits</i>. UAI. arXiv:2510.15483 H. Zenati, E. Diemert, M. Martin, J. Mairal, P. Gaillard (2023). <i>Sequential Counterfactual Risk Minimization</i>. ICML. arXiv:2302.12120

- F. El Khoury*, **H. Zenati***, N. Kallus, M. Arbel, A. Bibaut (2026). *Semiparametric Efficient Bilevel Gradient Estimation*. Preprint.
- B. Bozkurt, A. Galashov, D. Meunier, Z. Shen, A. Gretton, **H. Zenati** (2026). *Doubly Robust Proxy Causal Learning with Neural Mean Embeddings*. Preprint.
- Z. Shen, N. Kallus, D. Meunier, **H. Zenati**, A. Gretton, A. Bibaut (2026). *Nonparametric Instrumental Variable Analysis Without Structural Equations: Debiased Inference on Functionals of Inverse Problems with No Solutions*. Preprint.
- A. Bibaut, **H. Zenati**, T. Rahier, N. Kallus (2026). *Functional Natural Policy Gradients*. Preprint.
- H. Zenati**, B. Bozkurt, A. Gretton (2025). *Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings*. NeurIPS.
- B. Bozkurt, **H. Zenati**, D. Meunier, L. Xu, A. Gretton (2025). *Density Ratio-Free Doubly Robust Proxy Causal Learning*. NeurIPS.
- H. Zenati**, J. Abecassis, J. Josse, B. Thirion (2025). *Double Debiased Machine Learning for Mediation Analysis with Continuous Treatments*. AISTATS.
- J. Abecassis, **H. Zenati**, S. Boumaiza, J. Josse, B. Thirion (2025). *Causal Mediation Analysis with One or Multiple Mediators: A Comparative Study*. Psychological Methods.
- H. Zenati**, A. Bietti, M. Martin, E. Diemert, P. Gaillard, J. Mairal (2025). *Counterfactual Learning of Stochastic Policies with Continuous Actions*. Transactions on Machine Learning Research.
- J. Zhou, P. Gaillard, T. Rahier, **H. Zenati**, J. Arbel (2024). *Towards Efficient and Optimal Covariance-Adaptive Algorithms for Combinatorial Semi-Bandits*. NeurIPS.
- M. Martin, P. Mertikopoulos, T. Rahier, **H. Zenati** (2022). *Nested Bandits*. ICML.
- H. Zenati**, A. Bietti, E. Diemert, J. Mairal, M. Martin, P. Gaillard (2022). *Efficient Kernelized UCB for Contextual Bandits*. AISTATS.
- H. Zenati**, A. Bietti, M. Martin, E. Diemert, J. Mairal (2020). *Optimization Approaches for Counterfactual Risk Minimization with Continuous Actions*. ICLR Workshop on Causal Learning for Decision Making.
- P. Mertikopoulos, B. Lecouat, **H. Zenati**, C.-S. Foo, V. Chandrasekhar, G. Piliouras (2019). *Optimistic Mirror Descent in Saddle-Point Problems: Going the Extra (Gradient) Mile*. ICLR.
- K. Ouardini, H. Yang, B. Unnikrishnan, M. Romain, C. Garcin, **H. Zenati**, J. P. Campbell, M. F. Chiang, J. Kalpathy-Cramer, V. Chandrasekhar, et al. (2019). *Towards Practical Unsupervised Anomaly Detection on Retinal Images*. MICCAI Workshop on Domain Adaptation and Representation Transfer.
- H. Zenati**, M. Romain, C.-S. Foo, B. Lecouat, V. Chandrasekhar (2018). *Adversarially Learned Anomaly Detection*. ICDM.
- H. Zenati**, C.-S. Foo, B. Lecouat, G. Manek, V. R. Chandrasekhar (2018). *Efficient GAN-Based Anomaly Detection*. ICLR Workshop.
- B. Lecouat, C.-S. Foo, **H. Zenati**, V. Chandrasekhar (2018). *Semi-Supervised Learning with GANs: Revisiting Manifold Regularization*. ICLR Workshop.
- B. Lecouat, C.-S. Foo, **H. Zenati**, V. Chandrasekhar (2018). *Manifold Regularization with GANs for Semi-Supervised Learning*. arXiv preprint.
- B. Lecouat, K. Chang, C.-S. Foo, B. Unnikrishnan, J. M. Brown, **H. Zenati**, A. Beers, V. Chandrasekhar, J. Kalpathy-Cramer, P. Krishnaswamy (2018). *Semi-Supervised Deep Learning for Abnormality Classification in Retinal Images*. NeurIPS Workshop on Machine Learning for Health.

TEACHING	Data Visualisation , Université Paris Dauphine – PSL	2021–2022
	Instructor for Master’s students in the IASD program. Covered visualisation as a statistical and algorithmic tool for exploring, summarizing, and communicating complex data, with topics including graphical principles and dimensionality-reduction methods such as PCA, multidimensional scaling, and t-SNE.	
	Advanced Machine Learning , CentraleSupélec	2024–2025
	Teaching assistant alongside Emilie Chouzenoux. Helped students connect the formal ideas in modern machine learning, statistics, and optimization to practical algorithms and empirical behavior.	
	Causal Inference , New York University Paris	2024–2025
	Co-instructor with Judith Abecassis. Taught lectures and recitations on randomized and observational causal inference, including identification assumptions, weighting, matching, instrumental variables, and regression discontinuity.	
	Teaching philosophy: conceptual clarity, mathematical rigor, and active engagement.	
STUDENT	- Julien Zhou, Master’s student	2023
SUPERVISION	- Samuel Girard, PhD student	2025–2026
	- Robin Leman, Master’s student	2026
TALKS	- <i>Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings</i> , INRIA Thoth, France	Oct. 2025
	- <i>Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings</i> , Symposium on Mathematical Foundations of Trustworthy Learning, Switzerland	Oct. 2025
	- <i>Sequential Counterfactual Risk Minimization</i> , ELLIS Unconference, France	Jul. 2023
	- <i>Sequential Counterfactual Risk Minimization</i> , INRIA Lille, France	Jun. 2023
	- <i>Nested Bandits</i> , ICML, United States	Jul. 2022
	- <i>Efficient Kernelized Contextual Bandits</i> , virtual talk	May 2022
	- <i>Optimization Approaches for Counterfactual Risk Minimization with Continuous Actions</i> , virtual talk	Apr. 2020
	- <i>Adversarially Learned Anomaly Detection</i> , ICDM, Singapore	Nov. 2018
	- <i>Efficient GAN-Based Anomaly Detection</i> , DL2.0 Workshop, I2R, ASTAR, Singapore	Apr. 2018
SERVICE	Reviewing	
	AISTATS, ICML, NeurIPS, ICLR, TMLR	
	Reading Group Organization	
	Co-organized a reading group at Gatsby, UCL on van der Vaart’s <i>Asymptotic Statistics</i>	2026
	Organized a reading group at Inria Paris-Saclay on Wainwright’s <i>High-Dimensional Statistics</i>	2023–2024
	Workshop and Seminar Organization	
	Advances on Adaptive Experimentation (AAE) Workshop, Gatsby Unit	2026
	External machine learning seminars, Gatsby Unit, UCL	since Sept. 2025
SOFTWARE	med_bench : Python package for causal mediation analysis with a unified implementation of common estimators and flexible nuisance models.	
	Open-source research code:	
	• adaptive-KTE • counterfactual-policy-mean-embedding • double-debiased-machine-learning-mediation-continuous-treatments • sequential-counterfactual-risk-minimization • Efficient-Kernel-UCB • Nested-Exponential-Weights • optimization-continuous-action-crm • Adversarially-Learned-Anomaly-Detection • Efficient-GAN-Anomaly-Detection	